

HGCLIP: Exploring Vision-Language Models with Graph Representations for Hierarchical Understanding

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Abstract

Object categories are typically organized into a multi-granularity taxonomic hierarchy. When classifying categories at different hierarchy levels, traditional uni-modal approaches focus primarily on image features, revealing limitations in complex scenarios. Recent studies integrating Vision-Language Models (VLMs) with class hierarchies have shown promise, yet they fall short of fully exploiting the hierarchical relationships. These efforts are constrained by their inability to perform effectively across varied granularity of categories. To tackle this issue, we propose a novel framework (HGCLIP) that effectively combines CLIP with a deeper exploitation of the Hierarchical class structure via Graph representation learning. We explore constructing the class hierarchy into a graph, with its nodes representing the textual or image features of each category. After passing through a graph encoder, the textual features incorporate hierarchical structure information, while the image features emphasize class-aware features derived from prototypes through the attention mechanism. Our approach demonstrates significant improvements on 11 diverse visual recognition benchmarks. Our codes are fully available at <https://github.com/richard-peng-xia/HGCLIP>.

1 Introduction

Hierarchical image classification (Salakhutdinov et al., 2011; Guo et al., 2018) aims to enhance classification accuracy by identifying objects at various levels of granularity and capturing subtle relationships among them. Specifically, all the classes are organized into a multi-granularity taxonomic hierarchy (see in Figure 1a), where the top-level nodes represent broader categories (“Mammal”), while the lower-level nodes encompass finer-grained subcategories (“Dog”). The inherently hierarchical nature of the task compounds its complexity, as models must exhibit a keen understanding

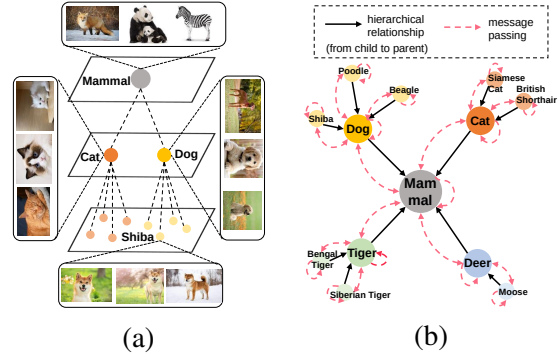


Figure 1: An illustration of the graph representation based on class hierarchy. (a) The class hierarchy is presented in a tree structure. (b) The hierarchical labels are constructed into a graph, with nodes representing the text/image features of each class. The graph is fed into a graph encoder, where the nodes update the parameters by aggregating the messages from their neighboring nodes. Thus, the class features are fused with hierarchical information via graph representation learning.

of semantic hierarchies, balancing the trade-off between capturing fine-grained details for subclasses while maintaining a broad understanding of super-classes (Chen et al., 2018). Previous works (Chang et al., 2021; Guo et al., 2018) mainly focus on enhancing image features according to the hierarchy of multiple branch outputs. These uni-modal methods only focus on the image modality, leading to certain limitations in complex scenarios, such as the inability to effectively utilize the textual descriptions of hierarchical labels and adapt to new classes or datasets. Therefore, leveraging multi-modal models (e.g., VLMs) to address hierarchical image classification presents stronger potential, offering richer information and greater scalability.

Given the powerful generalization capabilities of VLMs (Radford et al., 2021; Jia et al., 2021; Zhai et al., 2022) demonstrated on downstream tasks, harnessing their capabilities to address hierarchical image classification tasks presents a highly valuable exploration. These models are pre-trained on large-scale text-image pairs to align features from

the image and text modalities in a shared latent embedding space. The predicted probabilities are obtained by calculating the similarity between image features and text features.

Recently, some works have explored improving accuracy based on VLMs via class hierarchies. Specifically, CHiLS (Novack et al., 2023) employs hierarchical mappings to transform each class into a list of subcategories. However, this approach has significant drawbacks when applied to fine-grained datasets, as the subcategories of these labels tend to be specialized and rare, resulting in an overly detailed and contextually sparse representation. Utilizing these specific labels as prompts may overwhelm the model, lacking broader contextual relevance. Hierarchy-CLIP (Ge et al., 2023) proposes a label augmentation method that leverages the WordNet (Fellbaum, 2010) label hierarchy, enriching each class with its parent and child classes. This method aims to provide a richer semantic expansion of class descriptions. It enhances surface-level semantic associations rather than delving into the deeper and more structured connections inherent in a hierarchical structure. This limitation becomes apparent in scenarios requiring classification across multiple hierarchical levels, where a nuanced understanding of these relationships is crucial. Moreover, these methods are both training-free. While this offers the advantage of simplicity and direct application, it lacks the capacity for further model adaptation to specific datasets. Additionally, these methods do not fully exploit the potential of VLMs to adapt to the diverse and complex nature of hierarchical understanding.

Hence, the limitations of these approaches give rise to a new question: *How can models leverage the class hierarchy thoroughly to simultaneously improve the prediction accuracy of categories at different semantic granularity levels?*

To address this issue, we first introduce prompt learning (Zhou et al., 2022; Khattak et al., 2023a) as an efficient method to adapt VLMs to downstream tasks. HGCLIP introduces prompt tokens within the multi-modal branches of CLIP to facilitate the learning of hierarchical contextual representations. More importantly, as demonstrated in Figure 1b, HGCLIP explores the integration of CLIP with graph representations for hierarchical image classification. Specifically, hierarchical relationships are modeled as a graph, given that they inherently form a tree-like structure. Based on this graph, we employ a graph encoder (Veličković et al., 2018a)

to encode text features, enabling them to incorporate hierarchical structural information. Moreover, since image features represent features of individual patches/pixels rather than categories, we utilize prototype learning to represent image features for each category. Similarly, a graph encoder is leveraged to allow the prototypes to learn hierarchical relationships, and subsequently utilize the attention mechanism to enable the spatial feature map of images to focus more on the class-aware features derived from prototypes. On hierarchical image classification, HGCLIP outperforms existing CLIP-based approaches across both generic and fine-grained datasets. In scenarios where hierarchical labels are unavailable, HGCLIP also improves accuracy when utilizing class hierarchies queried by ChatGPT (OpenAI, 2023). Further, HGCLIP demonstrates favorable generalization ability and robustness in domain generalization and subpopulation shift settings, resulting in consistent improvements over existing methods. To sum up, the main contributions of this work include:

- We propose HGCLIP, a state-of-the-art (SoTA) method in hierarchical image classification for adaptation of CLIP.
- To better utilize label hierarchies, we explore the graph representations to incorporate hierarchical structural information into vision-language feature representations for effective hierarchical understanding.
- Our approach exhibits new SoTA performance across eleven hierarchical image classification benchmarks.

2 Related Work

Prompt Learning in Vision-Language Models:

VLMs leverage information from both image and text modalities to encode multimodal representations. VLMs, e.g., CLIP (Radford et al., 2021), ALIGN (Jia et al., 2021), and LiT (Zhai et al., 2022) are pre-trained on large-scale image-text pairs and demonstrate remarkable representation abilities on various downstream tasks (Gao et al., 2023; Zhu et al., 2022; Ding et al., 2022; Xia et al., 2024c,e,d,a). However, efficiently adapting them to downstream tasks is still a major challenge. Prompt learning (Li and Liang, 2021; Lester et al., 2021; Jin et al., 2024a,b,c), as a parameter-efficient technique, is well-suited for utilizing the representation capacity of pre-trained VLMs to boost performance, instead of the resource-intensive process

of full fine-tuning. Many works (Shu et al., 2022; Zhou et al., 2023; Li et al., 2024) have demonstrated powerful performance on specific downstream tasks by combining VLMs and prompt tuning.

Hierarchical Image Classification: Hierarchical image classification (Salakhutdinov et al., 2011; Guo et al., 2018) aims to categorize images into a hierarchical structure that organizes classes into a tree-like taxonomy. It acknowledges the inherent hierarchical nature of visual concepts, allowing for more nuanced and contextually rich image categorization. Prior research has explored various methodologies, including model architectures tailored for hierarchical classification (Guo et al., 2018; Chang et al., 2021; Chen et al., 2018), and exploiting the relationship of the categories in the hierarchy (Liu et al., 2022). Furthermore, the development of hierarchical classification has spurred some works (Ju et al., 2023, 2024; Yi et al., 2022) that harness the class hierarchy across diverse domains, as these models tend to focus more on fine-grained and semantically relevant features. Recently, hierarchical labels are integrated with VLMs (Novack et al., 2023; Ge et al., 2023). Nonetheless, these methods roughly overlook the hierarchical relationships among labels. Our work *comprehensively* leverages the hierarchical relationships among labels, resulting in performance improvements on *both generic and fine-grained* datasets.

Graph Representation Learning: Modern graph analysis methods rely on graph representation learning, encompassing graph embedding, graph neural networks (GNNs), and transformers. Early graph embedding techniques (Perozzi et al., 2014; Grover and Leskovec, 2016) typically map nodes into a low-dimensional space, capturing structural information. Recently, GNNs (Kipf and Welling, 2016; Veličković et al., 2018a) have become the mainstream technique in graph representation learning. They rely on a message-passing framework where each node refines its representation by recursively aggregating messages from its neighbors. Moreover, some recent approaches have also explored transformer-based architectures (Yun et al., 2019; Hu et al., 2020). Furthermore, the boom of graph representation learning also advances the research and development in other communities such as CV (Shi et al., 2019) and NLP (Zhang et al., 2023). In this work, we employ *hierarchical graph representations* to enrich multi-modal features, thus improving the model performance and

generalization.

3 Preliminaries

In this work, our goal is to learn hierarchical multi-modal knowledge via graph encoder based on CLIP. We will introduce related concepts and definitions in the following.

3.1 Revisiting CLIP

We denote the CLIP image and text encoder as $\mathcal{I}(\cdot)$ and $\mathcal{T}(\cdot)$. The dataset contains K categories, *i.e.*, $\{C_1, \dots, C_K\}$. CLIP leverages a structured approach by inserting all category names into a predefined textual template represented by the [CLASS] token, *e.g.*, creating expressions like “a photo of a [CLASS].”. This results in the generation of textual inputs denoted as T_K . Subsequently, textual features, represented as $\mathbf{F}_t \in \mathbb{R}^{K \times D}$, are extracted. Each input image I is divided into M fixed-sized patches, and each patch is embedded into D -dimensional latent space. Then CLIP derives its spatial feature map $\mathbf{F}_s \in \mathbb{R}^{H \times W \times D}$ and computes the global visual representations $\mathbf{f}_v \in \mathbb{R}^{1 \times D}$ through pooling operations, where H and W denote the height and width of the feature map. The integration of features from both encoders is achieved through cosine similarity measures, ultimately yielding classification *logits* $\in \mathbb{R}^{1 \times K}$. This comprehensive process can be summarized as follows

$$\mathbf{F}_t = \mathcal{T}(T_K), \quad (1)$$

$$\mathbf{f}_v = \text{POOLING}(\mathbf{F}_s), \quad \mathbf{F}_s = \mathcal{I}(I), \quad (2)$$

$$\text{logits} = \mathbf{f}_v \mathbf{F}_t^T. \quad (3)$$

The matrix multiplication operation between $\mathbf{f}_v$ and $\mathbf{F}_t$ is equivalent to calculating cosine similarities, assumed to be L_2 -normalized features. *logits* signifies the computed probabilities for all K categories, and CLIP identifies the category with the maximum output probability $\text{argmax}_{C_K}(\text{logits})$ as its final prediction.

3.2 Graph Encoder

Graph. A graph is represented as $G = (V, E)$, with V denoting the set of nodes and E the set of edges. Equivalently, the graph can be represented by an adjacency matrix A , such as $A_{ij} = 1$, if $(v_i, v_j) \in E$, for any $v_i, v_j \in V$.

Graph Encoder. GNNs are popular choices of graph encoder, most of which employ a message-passing mechanism (Wu et al., 2020). Specifically,

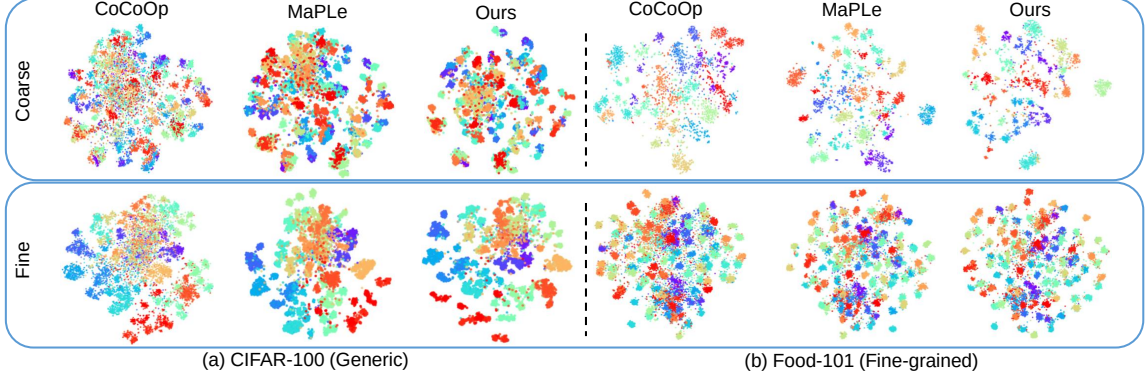


Figure 2: t-SNE plots of image embeddings in SOTA method CoCoOp, MaPLe, and HGCLIP on two datasets with distinct semantic granularities. HGCLIP shows better separability in both fine-grained and coarse-grained levels.

each node in the graph aggregates messages (*i.e.*, input features or embeddings) from its neighboring nodes to update its own embedding. Multiple layers of neighborhood aggregation can be stacked, facilitating recursive message passing across the graph. Formally, in the l -th GNN layer, the embedding of node v , denoted by $\mathbf{f}_v^l$, is calculated based on the embeddings in the previous layer, as follows

$$\mathbf{f}_v^l = \text{AGGR}(\mathbf{f}_v^{l-1}, \{\mathbf{f}_u^{l-1} : u \in \mathcal{N}_v\}; \theta^l), \quad (4)$$

where $\mathcal{N}_v$ is the set of neighboring nodes of v , θ^l is the learnable GNN parameters in layer l . $\text{AGGR}(\cdot)$ is the neighborhood aggregation function and can take various forms, ranging from the simple mean pooling (Kipf and Welling, 2016) to advanced neural networks such as neural attention (Veličković et al., 2018a) or multi-layer perceptrons (Xu et al., 2019). Note that the initial node embedding $\mathbf{f}_v^0$ is simply given by the input feature. We abstract the multi-layer encoding process as

$$\mathbf{f}_v = \text{GRAPHENCODER}(\mathbf{f}_v^0, \mathcal{N}_v; \Theta), \quad (5)$$

where $\Theta = (\theta^1, \dots, \theta^L)$ is the collection of weights across the layers. Note that graph embedding methods (Perozzi et al., 2014; Tang et al., 2015; Grover and Leskovec, 2016) and graph transformers (Yun et al., 2019; Hu et al., 2020; Ying et al., 2021) could also serve as GRAPHENCODER.

4 Methodology

In this section, as shown in Figure 3, we present our proposed method, *i.e.*, HGCLIP for adapting pre-trained VLMs for hierarchical understanding. Our approach aims to enhance the capacity for understanding multiple semantic levels. Most prior approaches focus on single-label classification, whereas hierarchical classification necessitates that the model attends to features relevant

to multi-granularity hierarchies. To this end, HGCLIP entails: **a)** introducing learnable prompt tokens within multiple transformer blocks in both the visual and textual branches to learn hierarchical contextual representations; **b)** employing a graph encoder to encode textual features, integrating them with hierarchical structural information; **c)** utilizing prototype learning to represent image features of each category and similarly modeling them utilizing a graph encoder, thereafter employing the attention mechanism to enable the spatial feature map of images to focus more on class-aware and hierarchy-guided image features.

4.1 Hierarchy Setting

The ground truth class hierarchy currently available in a dataset is usually obtained by querying a WordNet (Fellbaum, 2010)-like dictionary, but in the real world, our dataset may have no available class hierarchy. In this case, we turn to LLMs, *i.e.*, ChatGPT, to approximate the hierarchy diagram. Specifically, given some label set size K , semantic granularity levels h , class names, and optional context, we query ChatGPT with the prompt:

Generate h -tier hierarchical labels for the following K categories: $\{C_1, \dots, C_K\}$.

4.2 Multi-modal Hierarchical Prompt

In order to comprehensively and efficiently leverage the capabilities of pretrained VLMs, we explore the potential of multi-modal prompt, encompassing both textual and visual prompt. As highlighted in (Khattak et al., 2023a), the acquisition of prompt at deeper transformer layers is crucial, as it progressively models hierarchical feature representations. Learnable tokens are introduced at multiple transformer blocks of both textual and visual branches of VLMs, given as textual

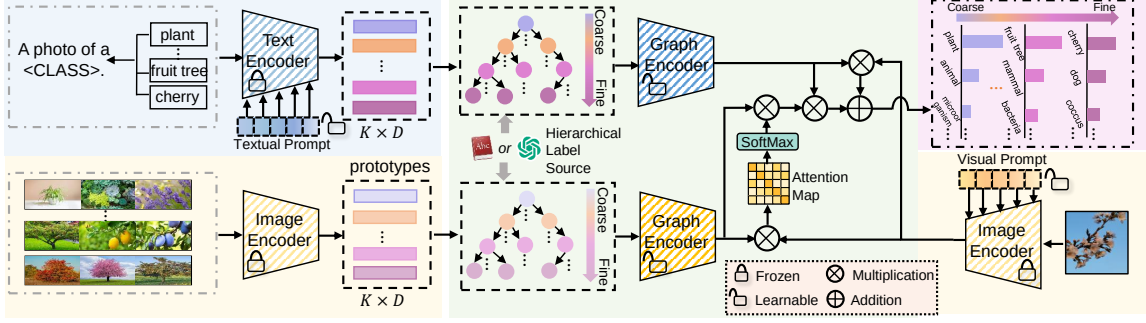


Figure 3: The pipeline of HGCLIP for adapting CLIP to hierarchical image classification. We introduce multi-modal hierarchical prompt to learn contextual representations. Then we construct the label hierarchy into a graph, with its nodes representing the textual or image features of each class. Features integrate hierarchical structure information through message passing in the graph encoder. Textual features directly combine hierarchical representations, while image features focus on class-aware prototypes through the attention mechanism.

prompt $\mathbf{P}^T = \{\mathbf{p}_1^T, \dots, \mathbf{p}_t^T\}$ and visual prompt $\mathbf{P}^V = \{\mathbf{p}_1^V, \dots, \mathbf{p}_v^V\}$, respectively. Therefore, the image encoder processes the input tokens added visual prompt $\mathbf{P}^V$ to generate prompted spatial feature map represented as $\tilde{\mathbf{F}}_s \in \mathbb{R}^{(HW+v) \times D}$ and prompted global visual representations $\tilde{\mathbf{f}}_v \in \mathbb{R}^{1 \times D}$. Similarly, textual prompt $\mathbf{P}^T$ are incorporated into the input tokens for encoding, and textual features are obtained as $\tilde{\mathbf{F}}_t \in \mathbb{R}^{K \times D}$. These hierarchical prompt tokens leverage the knowledge encoding capabilities of VLMs to effectively learn task-relevant contextual representations across different semantic levels.

4.3 Delving into Graph Representations

The hierarchical structure among labels naturally forms a tree structure, hence we leverage graph representations to model the hierarchy and integrate it into multi-modal features. In Figure 2, we visualize and compare the image embeddings of HGCLIP with those of previous SoTA CoCoOp and MaPLe. It is worth noting that the image embeddings of CLIP, CoOp, CoCoOp, and KgCoOp would be identical, as they do not learn prompts in the visual branch. The visualization reveals that the image embeddings of HGCLIP are more separable, indicating that incorporating hierarchical information can better adapt CLIP.

Encoding Text: Clearly, textual features $\tilde{\mathbf{F}}_t = \{\tilde{\mathbf{f}}_n^t\}_{n=1}^K$ can be directly employed as input for a graph encoder, as they possess corresponding D -dimensional textual features for each category. The class hierarchy is constructed into a graph, where vertices and edges represent individual classes and pairs of classes with hierarchical relationships, respectively. As a result, each node n of the text-attributed graph is associated with text features

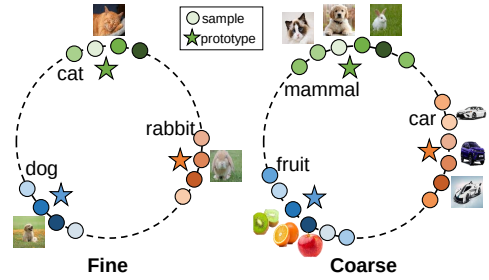


Figure 4: Semantic prototypes are constructed to guide the learning of hierarchical semantics of images.

of the corresponding category $\tilde{\mathbf{f}}_n^t$. The graph encoder approaches node classification by using the structural interactions between nodes. The textual features $\tilde{\mathbf{F}}_t = \{\tilde{\mathbf{f}}_n^t\}_{n=1}^K$ integrating hierarchical information are encoded as follows:

$$\{\hat{\mathbf{f}}_n^t\}_{n=1}^K = \text{GRAPHENCODER}(\tilde{\mathbf{f}}_n^t, \mathcal{N}_n; \Theta_t), \quad (6)$$

where Θ_t denotes the parameters of the graph encoder for textual modality, $\mathcal{N}_n$ denotes the neighbor nodes of n .

Encoding Image: In contrast to textual features, the spatial feature map represents the features of each patch, and the global visual representations characterize the image holistically, rather than representing features for each category. Therefore, the image features of each image cannot be directly input into the graph encoder. To address this issue, as shown in Figure 4, we first leverage prototype learning to represent the image features for each category. The global features $\mathbf{F}_v^* = \{\mathbf{f}_n^{v*}\}_{n=1}^K \in \mathbb{R}^{K \times 1 \times D}$ of all images $\{I_n\}_{n=1}^K$ (only in the training set) belonging to each class are extracted as prototypes for all categories. These prototypes can then be utilized by the graph encoder to be encoded. The procedure is as follows

$$\mathbf{F}_v^* = \text{POOLING}(\mathbf{F}_s^*), \quad \mathbf{F}_s^* = \mathcal{I}(I_K). \quad (7)$$

In the image-attributed graph, each node n is associated with image features $\{\mathbf{f}_n^{v*}\}_{n=1}^K$, while the rest is consistent with the text-attributed graph. Similarly, the image features $\mathbf{F}_{v*}$ are encoded as follows

$$\{\hat{\mathbf{f}}_{v*}^n\}_{n=1}^K = \text{GRAPHENCODER}(\mathbf{f}_{v*}, \mathcal{N}_n; \Theta_v), \quad (8)$$

where Θ_v denotes the parameters of the graph encoder for visual modality. After the visual graph encoder effectively leverages structural knowledge, we then employ the attention mechanism to obtain the attention weights of visual features $\tilde{\mathbf{F}}_s$ with respect to the prototypes $\hat{\mathbf{F}}_{v*}$. The calculation of attention weights is as follows

$$\psi = \tilde{\mathbf{F}}_s \hat{\mathbf{F}}_{v*}^T \in \mathbb{R}^{(HW+v) \times K}, \quad (9)$$

where ψ denotes the attention map. Each element of ψ represents the attention weight, namely, the feature similarity between a class prototype and one image pixel/site. Based on ψ , we update the spatial feature map as follows

$$\hat{\mathbf{F}}_s = \text{SoftMax}(\psi/\alpha) \tilde{\mathbf{F}}_s, \quad (10)$$

where α modulates the attention magnitude. Weighted by the attention scores representing similarity, the image features incorporate structured information from the prototypes. As the prototypes $\hat{\mathbf{F}}_{v*}$ encode K -category visual knowledge, the signals of classes appearing in the image would be more notable. Meanwhile, the spatial feature map provides pixel-level fine-grained information for the interaction, contributing to the thorough integration of class-aware features from the prototypes into the image features.

Classification Logits: Finally, we obtain the attention-interacted global visual feature by pooling and output the classification logits as

$$\hat{\mathbf{f}}_v = \text{POOLING}(\hat{\mathbf{F}}_s) \in \mathbb{R}^{1 \times D}, \quad (11)$$

$$\text{logits} = \lambda_1 \cdot \tilde{\mathbf{f}}_v \hat{\mathbf{F}}_t^T + \lambda_2 \cdot \hat{\mathbf{f}}_v \hat{\mathbf{F}}_t^T, \quad (12)$$

where λ_1 and λ_2 denote hyper-parameters to control the weight assigned to the *logits* that incorporate structured image features.

For hierarchical image classification, the model is required to simultaneously predict several labels at different granularities. Consequently, it is necessary to partition the predicted *logits* into their respective hierarchical categories logits_i , with each level corresponding to the ground truth labels GT_i ,

where $i = 1, \dots, h$. The overall loss function can be defined as follows

$$\mathcal{L} = \sum_{i=1}^h w_i \cdot \mathcal{L}_{CE}(GT_i, \text{logits}_i), \quad (13)$$

where w_i denotes the weights for learning features at different hierarchical levels and $\mathcal{L}_{CE}(\cdot, \cdot)$ represents a cross-entropy loss. A higher w_i prioritizes the learning of features at the i -th level, and vice versa.

5 Experiment

5.1 Benchmark Setting

Hierarchical Image Classification: We consider 11 visual classification datasets, covering a wide range of recognition tasks. These include two general object datasets, CIFAR-100 (Krizhevsky et al., 2009) and Caltech-101 (Fei-Fei et al., 2004); six fine-grained datasets, FGVC-Aircraft (Maji et al., 2013), StanfordCars (Krause et al., 2013), Food-101 (Bossard et al., 2014), Fruits-360 (Mureşan and Oltean, 2017), OxfordPets-37 (Parkhi et al., 2012) and ETHEC (Dhall et al., 2020); a scene recognition dataset SUN397 (Xiao et al., 2010); a texture dataset DTD (Cimpoi et al., 2014) and a satellite image dataset EuroSAT (Helber et al., 2019). The aim is to demonstrate our method under general situations of data diversity, where the label hierarchical levels range from two to four.

Domain Generalization: We evaluate the robustness of HGCLIP on out-of-distribution datasets (Xia et al., 2024b; Hu et al., 2024). The source distributions correspond to the original ImageNet (Deng et al., 2009). The task is to classify images from the target datasets (ImageNetV2 (Recht et al., 2019), ImageNet-Sketch (Wang et al., 2019), ImageNet-A (Hendrycks et al., 2021b) and ImageNet-R (Hendrycks et al., 2021a)), which consist of images that contain various types of domain shifts.

Implementation Details: We use top-1 accuracy to evaluate the prediction performance. We adopt CLIP ViT-B/16 as the visual encoder and use the corresponding CLIP Transformer as the text encoder. We set $\lambda_1 = 1$ and $\lambda_2 = 0.2$ to weight the proportion of hierarchical structural information. For hierarchical classification, we use deep prompting with $v = t = 4$ in the first 9 transformer layers and train the models for 50 epochs. All models are trained with a batch size of 64 and a learning rate

Dataset		CLIP ICML'21	CoOp IJCV'22	CoCoOp CVPR'22	VPT ECCV'22	MaPLe CVPR'23	KgCoOp CVPR'23	PromptSRC ICCV'23	HGCLIP (Ours)
CIFAR-100*	l_1	43.22	83.76	82.60	88.75	<u>90.67</u>	78.65	88.18	91.87
	l_2	66.57	76.81	75.73	83.94	<u>85.81</u>	73.49	82.24	86.55
Caltech-101	l_1	58.47	96.19	96.95	96.62	<u>98.06</u>	93.50	95.70	98.50
	l_2	69.01	95.12	94.18	95.06	<u>97.38</u>	93.56	95.57	97.51
	l_3	84.56	95.88	95.85	96.12	<u>96.88</u>	94.81	95.51	97.03
FGVC-Aircraft*	l_1	31.08	54.30	54.80	56.81	<u>70.79</u>	53.31	55.99	79.24
	l_2	35.49	51.59	50.42	53.00	<u>68.87</u>	50.38	50.20	70.70
	l_3	24.69	37.74	36.10	35.00	<u>52.58</u>	35.56	34.21	61.33
Stanford Cars*	l_1	61.75	82.39	83.31	83.46	83.35	82.78	82.85	83.61
	l_2	63.59	73.35	73.86	76.53	<u>76.92</u>	69.01	73.24	77.84
Food-101	l_1	61.38	87.04	89.13	89.28	88.16	85.18	87.37	91.12
	l_2	85.53	86.94	87.10	<u>88.46</u>	88.04	86.00	86.89	88.73
Fruits-360	l_1	75.43	93.73	99.21	99.46	<u>99.59</u>	90.95	94.41	99.71
	l_2	34.40	86.87	96.46	98.65	<u>98.31</u>	76.43	94.60	98.51
	l_3	23.55	85.90	96.58	97.12	97.78	72.80	92.13	<u>97.73</u>
Oxford Pets-37	l_1	99.97	99.86	99.86	<u>99.92</u>	99.91	99.89	99.89	<u>99.92</u>
	l_2	88.14	91.94	91.97	91.81	92.14	91.81	<u>92.19</u>	92.31
EuroSAT	l_1	62.97	91.50	91.77	92.41	<u>93.03</u>	91.38	91.96	95.57
	l_2	41.01	86.68	87.36	88.94	<u>90.16</u>	87.88	88.30	92.79
SUN397*	l_1	70.29	88.01	88.60	<u>90.82</u>	90.59	87.62	88.18	92.16
	l_2	63.59	84.28	84.20	86.29	86.72	85.11	85.40	88.39
	l_3	60.85	78.67	78.66	80.66	<u>81.37</u>	79.16	80.08	83.41
DTD	l_1	55.17	80.86	80.98	83.83	83.41	87.45	81.83	86.82
	l_2	48.09	74.60	72.19	<u>78.34</u>	<u>78.36</u>	75.68	75.79	81.08
ETHEC*	l_1	31.12	89.45	89.61	90.87	<u>92.17</u>	86.03	90.64	95.76
	l_2	2.65	85.10	86.07	86.79	<u>89.60</u>	83.11	87.91	93.40
	l_3	17.94	74.67	75.02	75.81	<u>78.48</u>	71.33	77.46	82.98
	l_4	1.52	49.48	51.27	51.99	55.73	47.62	<u>55.75</u>	60.39

Table 1: Top-1 accuracy (%) comparison on hierarchical image classification of HGCLIP with previous CLIP-based prompt tuning methods. The best result is **bold** and the second best is underlined. * denotes that the dataset is with available class hierarchy, and hierarchies of others are queried through ChatGPT. l_i represents the classification accuracy at the i -th hierarchical level, where a smaller i indicates a coarser granularity level, and vice versa.

of $3e-4$ via SGD optimizer, and decay by the cosine annealing rule during training.

5.2 Hierarchical Image Classification

CLIP-based prompt tuning methods. Table 1 displays the comparative performance of zero-shot CLIP, recent works on prompt learning and HGCLIP on 11 diverse hierarchical classification datasets. In the case of CLIP, we utilize handcrafted specific prompts designed for each dataset. In comparison with state-of-the-art MaPLe (Khattak et al., 2023a) and PromptSRC (Khattak et al., 2023b), HGCLIP exhibits improved performance across all levels on all the datasets, with the exception of a slight decline in performance on Fruits-360. With the contribution of graph representations, as opposed to SoTA MaPLe and PromptSRC, HGCLIP demonstrates superior generalization across multiple hierarchical categories on all the datasets, achieving an absolute average gain of 2.2% and 5.7% respectively.

CLIP-based feature adaptation methods. In Table 2, we compare HGCLIP with prior feature adaption methods based on CLIP. CLIP-Adapter (Gao et al., 2023) learns two residual-

style adapters after CLIP. Tip-Adapter (Zhang et al., 2022) constructs a key-value cache model by extracting features from few-shot data, then views the cache model as a well-performing initialization and fine-tunes the cache keys. CALIP (Guo et al., 2023) is proposed to boost CLIP performance via a parameter-free attention module between multi-modal representations. In comparison with these feature adaption approaches, HGCLIP exhibits excellent feature representation capabilities, with an accuracy on CIFAR-100 that is 8.7%, 6.2%, and 13.3% higher than theirs, respectively.

Visual-only hierarchical image classification methods. We have analysed various multi-modal methods above, and to demonstrate the effectiveness of HGCLIP, we compare visual-only fine-grained visual classification methods, as shown in Table 3. Our method still achieve a significant advantage. Additionally, the visual-only FGVC methods are more time-consuming compared to ours (100 v.s. 50 training epochs).

5.3 Distribution Shifts

Domain Generalization: Table 5 summarizes the results of HGCLIP and prior approaches on out-

Method	Acc. %	
	l_1	l_2
CLIP (Radford et al., 2021) ICML'21	43.22	66.57
Linear probe	75.60	71.27
CLIP-Adapter (Gao et al., 2023) IJCV'23	83.91	77.03
Tip-Adapter (Zhang et al., 2022) ECCV'22	84.57	81.42
CALIP (Guo et al., 2023) AAAI'23	77.51	74.28
HGCLIP (Ours)	91.87	86.55

Table 2: Comparison with CLIP-based feature adaption methods.

Method	Acc. %	
	l_1	l_2
PMG (Du et al., 2020) ECCV'20	87.16	83.02
FGN (Chang et al., 2021) CVPR'21	87.88	83.60
GHORD (Zhao et al., 2021) CVPR'21	87.93	84.36
CHRF (Liu et al., 2022) ECCV'22	88.67	84.91
TFGIC (Xu et al., 2023) AAAI'23	89.20	85.17
HGCLIP (Ours)	91.87	86.55

Table 3: Comparison with visual-only SOTA FGVC methods.

of-distribution datasets. We verify the transferability of models trained on ImageNet to various ImageNet variants with domain shifts. On the target datasets, the performance of HGCLIP surpasses previous SoTA methods. This achievement underscores the efficacy of hierarchical graph representations to enhance multi-modal features. Such an integration improves the generalization capabilities, enabling it to perform well across varying domains. This indicates that HGCLIP not only captures the intricate relationships within the data but also adapts effectively to new and unseen domains.

Subpopulation Shift: We also examine HGCLIP’s robustness to subpopulation shift within a dataset. The source and target distributions, though encompassing identical class categories, feature distinct subpopulations within those classes. Our empirical investigations are executed on the four BREEDS (Santurkar et al., 2021) ImageNet subsets: Living17, Nonliving26, Entity13, and En-

Module				CIFAR-100		FGVC-Aircraft		
TP	TG	VP	VG	l_1	l_2	l_1	l_2	l_3
×	×	×	×	43.22	66.57	31.08	35.49	24.69
✓	×	×	×	84.21	77.22	54.96	52.67	38.72
×	×	✓	×	84.21	77.22	56.81	53.00	35.00
✓	×	×	×	87.42	81.24	61.56	57.90	42.83
×	×	✓	✓	87.18	80.87	61.52	58.13	43.17
✓	×	✓	×	90.67	85.81	70.79	68.87	52.58
✓	×	✓	✓	91.28	86.04	74.61	69.27	55.66
✓	✓	✓	×	91.43	85.96	75.37	69.28	57.50
✓	✓	✓	✓	91.87	86.55	79.24	70.70	61.33

Table 4: Component Analysis of HGCLIP. TP and VP serve as textual and visual prompts. TG and VG denote graph encoder for textual and visual modality.

Method	Source	Target			
	ImageNet	-V2	-Sketch	-A	-R
CLIP	86.11	80.19	72.13	46.11	70.85
CoOp	85.44	79.71	70.85	45.63	70.91
CoCoOp	85.03	79.49	72.96	46.99	72.47
MaPLe	<u>91.22</u>	<u>85.12</u>	74.19	49.28	73.79
PromptSRC	89.89	83.79	75.04	49.37	74.22
HGCLIP	92.19	86.24	77.40	50.38	76.07

Table 5: Domain generalization. Comparison with CLIP-based methods on robustness to domain shifts.

tity30. We further evaluate the generalizability of HGCLIP on four datasets from BREEDS (subsets of ImageNet) that exhibit subpopulation shifts and provide available class hierarchies. Figure 5 depicts the performance of HGCLIP and previous methods under subpopulation shifts. The models are trained only on base classes and tested on novel classes. The results suggest that HGCLIP possesses strong generalization capabilities even when confronted with feature-level shifts, underscoring the efficacy of hierarchical structure information in enhancing model generalizability. This success demonstrates the robustness of HGCLIP in handling variations within subpopulations, ensuring consistent accuracy and reliability across diverse and shifting data landscapes.

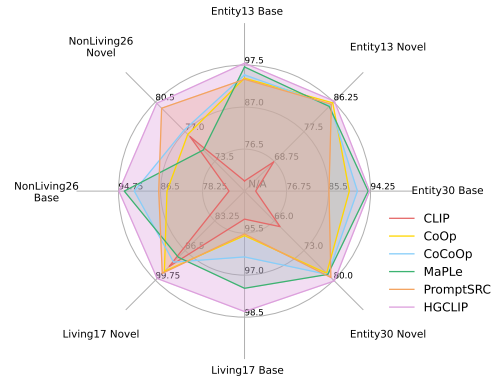


Figure 5: Results for different methods on the four BREEDS datasets (Santurkar et al., 2021) to measure robustness to subpopulation shift.

5.4 Ablative Analysis

Components Analysis. HGCLIP primarily consists of multi-modal prompts and graph encoders. In Table 4, we ablate on the performance of each module. The prompts facilitate the model in learning hierarchical contextual features, while the graph encoders effectively integrate hierarchical structure information into the feature representations. This enables the model to achieve impressive

results across multiple semantic granularities.

Noisy Hierarchies Queried by LLMs. It is important to note that LLMs may output sub-optimal hierarchical labels. LLMs produce inconsistent hierarchical labels based on a set of input category names or generate hierarchical labels of different levels, leading to certain biases in the model performance. However, even when utilizing the noisy hierarchy, HGCLIP still enhances accuracy within the original categories in the dataset.

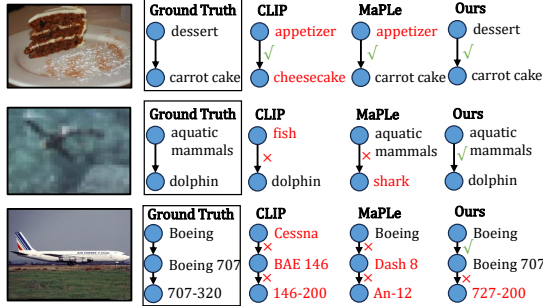


Figure 6: Example decisions from our model, MaPLE and CLIP.

5.5 Qualitative Analysis

Figure 6 presents illustrative cases showcasing the predicted probabilities of the models at different semantic granularity levels. CLIP shows inconsistencies in classification results at different levels, indicating that CLIP does not grasp the semantic relationship between different hierarchical levels. MaPLE improves prediction accuracy via learning hierarchical feature representation. However, it still displays inconsistencies when predicting classifications across different levels. Our method largely mitigates this issue, leveraging hierarchical graph representation to bolster the learning of inter-level class features.

5.6 Graph Encoder Analysis

We further conduct experiments to analyze the impact of various graph encoders. We apply three of the most commonly used graph learning models: GCN (Kipf and Welling, 2016), GAT (Veličković et al., 2018b), and GraphSAGE (Hamilton et al., 2017) to HGCLIP, the results are illustrated in Figure 7. First, for both hierarchical levels, GAT consistently exhibits superior performance, particularly at the fine-grained level, where GAT surpasses the other encoders. Second, with the increase of layer depth of the graph encoder, the accuracy initially rises. Upon reaching a peak (3 layers), the accuracy begins to gradually decline with further

increase in layer depth. Therefore, we use a 3-layer GAT as graph encoder in our experiments.

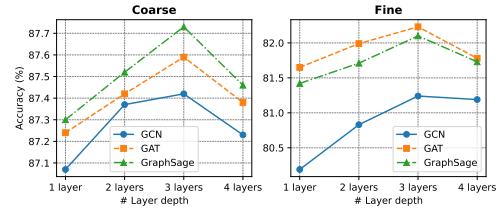


Figure 7: The impact of various GNN variants and the number of layers on hierarchical classification. We report accuracy results on two hierarchical levels.

6 Conclusion

In this work, we propose a novel view that combines VLMs with graph representations. Deep prompts are incorporated into the multi-modal branches, allowing VLMs to better learn hierarchical representations. The graph-based hierarchical relationships are encoded into features, strengthening the connection of features across multiple granularities. When integrating image features with graph representation, given that image features are pixel/region-level, prototype learning is employed for class-level image features, which are then fused with the image features through the attention mechanism. HGCLIP achieves SoTA results on several hierarchical image classification benchmarks.

Limitations

Although this work has achieved results by utilizing graph representations to characterize hierarchical information, it merely employs a simple graph encoder (such as GCN, GAT, GraphSAGE) to encode structural information. With the advancement of graph learning, it is anticipated that there will be better graph learning methods for representing hierarchical structures, which could further enhance features. We separately employ graph encoders for different modalities, yielding the best performance, outperforming cross-modal graph encoders with shared weights. This may be because current multi-modal models lack the ability to accurately extract features from each modality. With the development of multimodal models, this limitation is expected to be addressed, significantly reducing training time and improving inference speed. Additionally, we leverage multi-modal prompt learning but have not integrated it with graph learning, perhaps combining the two could yield better results.

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